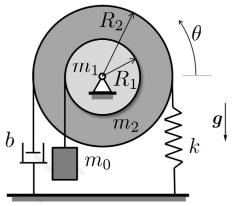
Problems Week 11

Problem 1: Composite Cylinder Vibration

Two uniform cylinders of masses m_1 and m_2 and radii R_1 and R_2 are welded together, as shown below. This composite cylinder rotates without friction about a hinge located at its center. An inextensible massless string is wrapped without slipping around the larger cylinder, whose two ends are connected to the ground through a spring of stiffness k and a damper of damping coefficient b. The smaller cylinder is connected to a weight m_0 via another inextensible massless string wrapped without slipping around the smaller cylinder. The weight is constrained to move vertically. Assume that the strings are able to sustain compressive forces, and let θ be a small angle. The spring is unstretched when $\theta = 0$.

Given: $m_0, m_1, m_2, R_1, R_2, k, b, |\theta(t)| \ll 1, g$



(a) What is the equation of motion of the system in terms of the rotation angle $\theta(t)$?

(b) At which rotation angle θ_{eq} is the system in static equilibrium?

(c) What is the natural frequency ω_0 and characteristic damping δ of the system?

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i^{\mathrm{nc}} \quad \text{with} \quad \mathcal{L} = T - V, \quad Q_i^{\mathrm{nc}} = \sum_{j=1}^{N} \boldsymbol{F}_j^{\text{non-cons.}} \cdot \frac{\partial \boldsymbol{r}_j}{\partial q_i}$$

a) (i) $T = \underbrace{\frac{1}{2} m_0 R_1^2 \dot\theta^2}_{\text{translation block}} + \underbrace{\frac{1}{2} \cdot \frac{m_1 R_1^2}{2} \cdot \dot\theta^2}_{\text{rotation disc 1}} + \underbrace{\frac{1}{2} \cdot \frac{m_2 R_2^2}{2} \cdot \dot\theta^2}_{\text{rotation disc 2}}$

(ii) $V = \underbrace{-m_0 g R_1 \theta}_{\text{mgh (block)}} + \underbrace{\frac{1}{2} k R_2^2 \theta^2}_{\substack{\text{spring potential} \\ (\Delta x = R_2\theta)}}$

$$\mathcal{L} = T - V = \frac{m_0 R_1^2 \dot\theta^2}{2} + \frac{m_1 R_1^2 \dot\theta^2}{4} + \frac{m_2 R_2^2 \dot\theta^2}{4} - \left(m_0 g R_1 \theta + \frac{1}{2} k R_2^2 \theta^2\right)$$

(iii) $\underline{Q}^{nc} = \underline{F}_d \cdot \frac{\partial \underline{r}}{\partial \theta}$ ← the only DoF in our sys.

$= c \cdot R_2 \dot\theta \underline{e}_2 \cdot (-R_2)$

$= -c R_2^2 \dot\theta$

$\underline{r} = R_2 \cos\theta \, \underline{e}_1 + R_2 \sin\theta \, \underline{e}_2$

$\frac{\partial \underline{r}}{\partial \theta} = R_2 \cos\theta \, \underline{e}_2 - R \sin\theta \, \underline{e}_1$

$\left.\frac{\partial \underline{r}}{\partial \theta}\right|_{\theta = \pi} = -R_2$

$$\frac{d}{dt}\left(m_0 R_1^2 \dot\theta + \frac{m_1 R_1^2 \dot\theta}{2} + \frac{m_2 R_2^2 \dot\theta}{2}\right) - \left(m_0 g R_1 - k R_2^2 \theta\right) = -c R_2^2 \dot\theta$$

$$\boxed{\left[\left(m_0 + \frac{m_1}{2}\right) R_1^2 + m_2 \frac{R_2^2}{2}\right] \ddot\theta + c R_2^2 \dot\theta + k R_2^2 \theta = m_0 g R_1}$$

b) $\frac{\partial V}{\partial q} = 0 \Rightarrow k R_2^2 \theta - m_0 g R_1 = 0 \Rightarrow \theta_{eq} = \frac{m_0 g R_1}{k R_2^2}$

c) $\omega_0^2 = \frac{k R_2^2}{\left(m_0 + \frac{m_1}{2}\right) R_1^2 + m_2 \frac{R_2^2}{2}}$

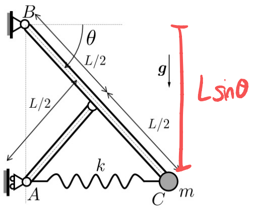
$$m\ddot{x} + c\dot{x} + kx = F(t) \quad \text{so} \quad \omega_0 = \sqrt{\frac{k}{m}}, \quad \delta = \frac{c}{2m}, \quad f(t) = \frac{F(t)}{m}$$

$\delta = \frac{c R_2^2}{(2m_0 + m_1) R_1^2 + m_2 R_2^2}$

Problem 2: Bar-Spring System I

Consider a system that consists of two bars of negligible masses and of lengths L and $L/2$. The bars connect three points A, B and C, as sketched below. A spring of stiffness k and unstretched length L_0 connects A and C. Furthermore, a particle of mass m is rigidly connected to point C. The system is hinged at B, while point A is constrained to move only in the vertical direction. Gravity acts downwards, as shown.

Given: m, L, k, g



(a) For what value of L_0 is $\theta = \pi/4$ an equilibrium position?

For the next task denote the unstretched spring length simply by L_0.

(b) For which values of k is $\theta = \pi/4$ a *stable* equilibrium position?

a) equilibrium $\Rightarrow \frac{\partial V}{\partial q} = 0$, where $V = -mgL\sin\theta + \frac{1}{2}k\underbrace{(L\cos\theta - L_0)}_{\Delta x}^2$

$\hookrightarrow \frac{\partial V}{\partial \theta} = -mgL\cos\theta_{eq} + k\cdot(L\cos\theta_{eq} - L_0)\cdot(-L\sin\theta_{eq}) = 0$

$$L_0 = \frac{mg\cos\theta_{eq}}{k\sin\theta_{eq}} + L\cos\theta_{eq}$$

$\to$ plug in $\frac{\pi}{4}$ $\Rightarrow L_0 = \frac{mg}{k} + \frac{\sqrt{2}}{2}L$

b) stable $\to \frac{\partial^2 V}{\partial \theta^2} > 0 \Rightarrow mgL\sin\theta + k\cdot L^2\sin^2\theta - kL\cos\theta(L\cos\theta - L_0) > 0$

$\to$ plug in $\frac{\pi}{4}$: $\frac{\sqrt{2}}{2}mgL + \frac{kL^2}{2} - kL\frac{\sqrt{2}}{2}\left(\frac{\sqrt{2}}{2}L - L_0\right) > 0$

$\frac{\sqrt{2}}{2}mgL + \frac{kL^2}{2} + \frac{\sqrt{2}}{2}kLL_0 > \frac{kL^2}{2}$

$\frac{\sqrt{2}}{2}mgL + \frac{\sqrt{2}}{2}kLL_0 > 0$

$mg + kL_0 > 0$ $\Rightarrow$ Satisfied for all $\underline{\underline{k > 0}}$!

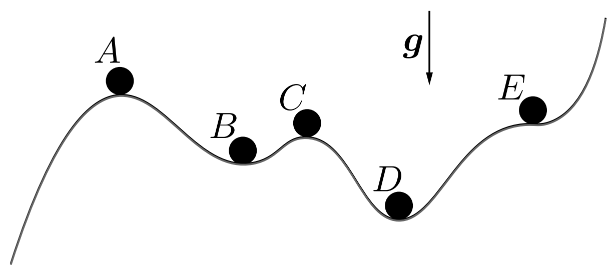
Exercise #11 (Solutions)

Lagrange Equations, Stability, Eigenfrequencies

November 28/29, 2024

Clicker 1: Particles on a hillside

A particle is placed at five different locations, labelled A through E, on a hillside, whose altitude profile is sketched below. Gravity acts downwards.



Which of the following statements is *not* correct?

(a) The potential energy V of a particle in this hillside has exactly four extrema (minima or maxima).

(b) Each position A through E is an equilibrium.

(c) Positions A, C, and E are metastable equilibria.

(d) Positions B and D are stable equilibria.

(e) Position E is a metastable equilibrium.

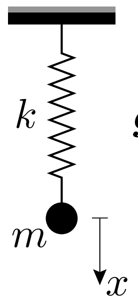
Solution:

By definition, an equilibrium is characterized by a vanishing derivative of the potential energy V, which includes potential energy minima, maxima and saddle points (here, positions A, B, C, D and E are all equilibria). A *stable* equilibrium is characterized by a (local or global) minimum of the potential energy – so positions B and D are stable equilibria. An *unstable* equilibrium is characterized by a (local or global) maximum of the potential energy – so positions A and C are unstable equilibria. Finally, a *metastable* equilibrium implies a saddle point of the potential energy – which here applies to position E. Based on the above, all statements are correct except for (c) because positions A and C are not metastable but unstable.

Clicker 2: Particle oscillation

A particle of mass m is attached to a linear spring of stiffness k, hanging vertically from the ceiling. Gravity is the only force acting (in the downward direction), as indicated.

<u>Given:</u> $m = 1\,\text{kg}$, $k = \pi^2 \frac{N}{m}$, $g = 9.81 \frac{m}{s^2}$



Which of the four diagrams below represent possible time histories $x(t)$ of the particle's position? (Time t is in seconds, position $x(t)$ in millimeter; and assume $x = 0$ is the equilibrium solution.)

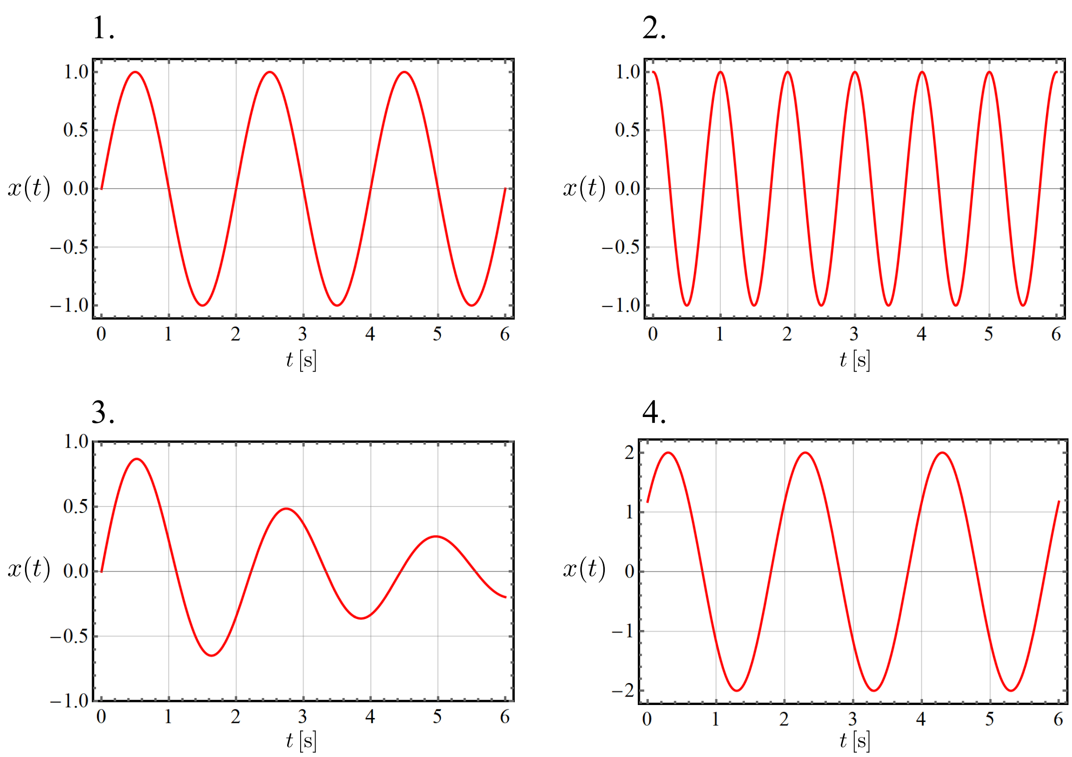
(a) All.

(b) None.

(c) All except 3.

(d) Only 1. and 2.

(e) Only 1. and 4.



Solution: Let us first derive the equations of motion of this system. We start by obtaining the kinetic and potential energies and the Lagrangian of the system as, respectively,

$$T = \frac{1}{2}m\dot{q}^2, \quad V = -mgq + \frac{1}{2}kq^2 \qquad \Rightarrow \qquad \mathcal{L} = T - V = \frac{1}{2}m\dot{q}^2 + mgq - \frac{1}{2}kq^2, \tag{1}$$

where we considered the only degree of freedom $q = x$, and define q such that the spring is unstretched when $q = 0$. The Lagrange's equation then evaluates to

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q} = m\ddot{q} - mg + kq \qquad \Leftrightarrow \qquad m\ddot{q} + kq = mg. \tag{2}$$

As seen in class, this equation of motion represents an undamped vibration, whose general solution is of the form

$$x(t) = A_1\cos(\omega_0 t) + A_2\sin(\omega_0 t) = A\sin(\omega_0 t + \varphi_0) \qquad \text{with} \quad A_i, A, \varphi_0 \in \mathbb{R} \tag{3}$$

and the eigenfrequency

$$\omega_0 = \sqrt{k/m} = \pi\,\mathrm{Hz}. \tag{4}$$

Coefficients A_1 and A_2 (or A and φ_0) depend only on the initial conditions and determine the constant amplitude of the vibration.

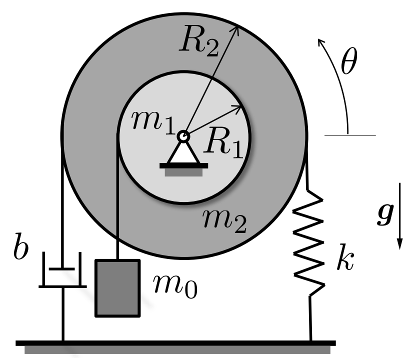
Considering the above solution of a free vibration, diagram 3. must be incorrect, since it shows a vibration whose amplitude decays over time, corresponding to a damped system. Furthermore, from the eigenfrequency ω_0 we obtain the time period of a complete vibration cycle as $T = 2\pi/\omega_0 = 2\,\mathrm{s}$. Since diagram 2. shows a vibration with a period of $T = 1$, it must also be incorrect. Lastly, both diagrams 1. and 4. show harmonic time responses with the correct period of $T = 2s$, and therefore correctly correspond to (3). Note that the differences between the two can stem from different initial conditions. Therefore, 1. and 4. are correct, so that response (e) is correct.

Note that the unstretched spring length as well as gravity has no effect on the eigenfrequency and only affects the equilibrium solution. Since $x(t)$ is measured with respect to the equilibrium solution, the response is independent of the unstretched length and g.

Problem 1: Composite cylinder vibration

Two uniform cylinders of masses m_1 and m_2 and radii R_1 and R_2 are welded together, as shown below. This composite cylinder rotates without friction about a hinge located at its center. An inextensible massless string is wrapped without slipping around the larger cylinder, whose two ends are connected to the ground through a spring of stiffness k and a damper of damping coefficient b. The smaller cylinder is connected to a weight m_0 via another inextensible massless string wrapped without slipping around the smaller cylinder. The weight is constrained to move vertically. Assume that the strings are able to sustain compressive forces, and let θ be a small angle. The spring is unstretched when $\theta = 0$.

Given: m_0, m_1, m_2, R_1, R_2, k, b, $|\theta(t)| \ll 1$, g



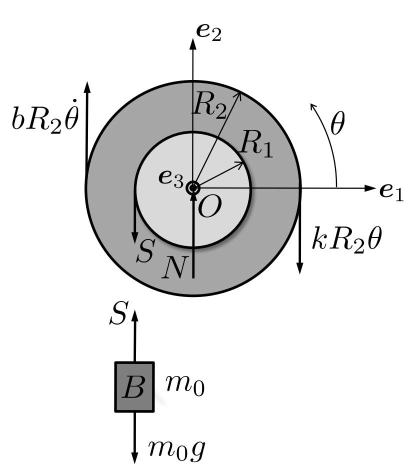
(a) What is the equation of motion of the system in terms of the rotation angle $\theta(t)$?

(b) At which rotation angle θ_{eq} is the system in static equilibrium?

(c) What is the natural frequency ω_0 and characteristic damping δ of the system?

Solution:

(a) We can use both the Lagrange equations and the linear and angular momentum balance (applied to the different subsystems) to obtain the sought equation of motion. To illustrate both ways, we will first consider Lagrange's equation and later verify our solution by re-deriving the equation of motion via the balance laws. Let us first draw the free-body diagram, as shown below:



Note that we can evaluate the distance travelled by the dashpot (and spring) and the weight by $R_2\theta$ and $R_1\theta$, respectively. To evaluate the Lagrange equation, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right)-\frac{\partial\mathcal{L}}{\partial\theta}=Q_{\mathrm{nc}},\quad \text{with } \mathcal{L}=T-V, \tag{5}$$

we must obtain the potential V and the kinetic energy T of the system. The kinetic energy writes

$$T=\frac{1}{2}\left(\frac{1}{2}m_1R_1^2\dot{\theta}^2+\frac{1}{2}m_2R_2^2\dot{\theta}^2+m_0R_1^2\dot{\theta}^2\right), \tag{6}$$

while the potential energy is given by

$$V=\frac{1}{2}kR_2^2\theta^2-m_0gR_1\theta. \tag{7}$$

We also need to consider the generalized force associated with the damper (see Eq. (4.8) in the Lecture Notes), which evaluates to

$$Q_{\mathrm{nc}}=\boldsymbol{F}_{\mathrm{d}}\cdot\frac{\partial\boldsymbol{r}}{\partial\theta}, \tag{8}$$

where $\boldsymbol{F}_{\mathrm{d}}$ is the force of the dashpot, given by $\boldsymbol{F}_{\mathrm{d}}=bR_2\dot{\theta}\boldsymbol{e}_2$, while $\boldsymbol{r}$ is the position vector of the point of application of the force. We thus need an expression for $\boldsymbol{r}$ in terms of θ to compute the derivative in (8). The position of a point on the edge of the outer disk (in $\boldsymbol{e}_1$-direction for $\theta=0$) is given by

$$\boldsymbol{r}=R_2\cos\theta\boldsymbol{e}_1+R_2\sin\theta\boldsymbol{e}_2, \tag{9}$$

and thus

$$\frac{\partial\boldsymbol{r}}{\partial\theta}=-R_2\sin\theta\boldsymbol{e}_1+R_2\cos\theta\boldsymbol{e}_2. \tag{10}$$

By substituting $\theta=\pi$ (which corresponds to the point of application of the damper force), (10) becomes

$$\left.\frac{\partial\boldsymbol{r}}{\partial\theta}\right|_{\theta=\pi}=-R_2\boldsymbol{e}_2. \tag{11}$$

Therefore, we arrive at

$$Q_{\mathrm{nc}}=\boldsymbol{F}_{\mathrm{d}}\cdot\frac{\partial\boldsymbol{r}}{\partial\theta}=-bR_2^2\dot{\theta}. \tag{12}$$

The equation of motion is then derived by computing each term of

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right)-\frac{\partial\mathcal{L}}{\partial\theta}=-bR_2^2\dot{\theta}. \tag{13}$$

By insertion we obtain the following non-zero derivatives

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial\dot{\theta}} = \left[\left(m_0+\frac{m_1}{2}\right)R_1^2+m_2\frac{R_2^2}{2}\right]\ddot{\theta}, \tag{14}$$

$$\frac{\partial V}{\partial\theta} = kR_2^2\theta-m_0gR_1, \tag{15}$$

and thus (collecting all terms depending on θ on one side)

$$\left[\left(m_0+\frac{m_1}{2}\right)R_1^2+m_2\frac{R_2^2}{2}\right]\ddot{\theta}+bR_2^2\dot{\theta}+kR_2^2\theta=m_0gR_1. \tag{16}$$

This is the equation of motion for $\theta(t)$.

Alternative solution using balance laws: By evaluating the angular momentum of the two disks with respect to point O it is possible to avoid computing the reaction forces of the hinge. Since this problem reduces to 2D, the angular momentum is then given by

$$M_O = I_O \dot{\omega} \tag{17}$$

where I_O is the moment of inertia of the two disks, i.e.,

$$I_O = \frac{1}{2}(m_1 R_1^2 + m_2 R_2^2). \tag{18}$$

The external torque is composed by the spring, dashpot and string torque, i.e.,

$$M_O = -kR_2^2\theta - bR_2^2\dot{\theta} + R_1 S. \tag{19}$$

Note that we can solve for the force S of the string by applying the linear momentum balance with respect to the weight, which gives

$$S = m_0 g - m_0 R_1 \ddot{\theta}. \tag{20}$$

Substituting the previous expressions into (17) then gives us (with $\dot{\omega} = \ddot{\theta}$)

$$-kR_2^2\theta - bR_2^2\dot{\theta} + m_0 g R_1 - m_0 R_1^2\ddot{\theta} = \frac{1}{2}(m_1 R_1^2 + m_2 R_2^2)\ddot{\theta}. \tag{21}$$

After readjusting the terms, we indeed obtain the identical form of the equation of motion as derived using the Lagrange equation, i.e.,

$$\left[\left(m_0 + \frac{m_1}{2}\right) R_1^2 + m_2 \frac{R_2^2}{2}\right]\ddot{\theta} + bR_2^2\dot{\theta} + kR_2^2\theta = m_0 g R_1. \tag{22}$$

(b) The rotation angle θ_{eq}, at which the system is in static equilibrium, can be obtained by solving

$$\frac{\partial V}{\partial \theta} = kR_2^2\theta - m_0 g R_1 = 0, \tag{23}$$

where we neglected Q_{nc} (since $\dot{\theta} = 0$ in the static equilibrium), which gives

$$\theta_{\text{eq}} = \frac{m_0 g R_1}{kR_2^2}. \tag{24}$$

(c) The natural frequency of the system is obtained from the equation of motion derived above (as the ratio of the terms in front of $\ddot{\theta}$ and θ, see the Formula Collection) and is given by

$${\omega_0}^2 = \frac{kR_2^2}{\left(m_0 + \frac{m_1}{2}\right) R_1^2 + m_2 \frac{R_2^2}{2}}, \tag{25}$$

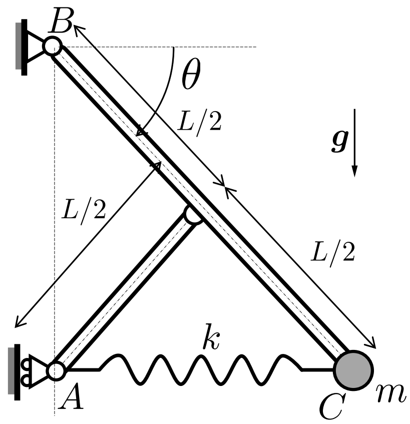
while the characteristic damping is half the ratio of those terms in front of $\dot{\theta}$ and $\ddot{\theta}$, so

$$\delta = \frac{bR_2^2}{2\left[\left(m_0 + \frac{m_1}{2}\right) R_1^2 + m_2 \frac{R_2^2}{2}\right]}. \tag{26}$$

Problem 2: Bar-spring system I

Consider a system that consists of two bars of negligible masses and of lengths L and $L/2$. The bars connect three points A, B and C, as sketched below. A spring of stiffness k and unstretched length L_0 connects A and C. Furthermore, a particle of mass m is rigidly connected to point C. The system is hinged at B, while point A is constrained to move only in the vertical direction. Gravity acts downwards, as shown.

Given: m, L, k, g



(a) For what value of L_0 is $\theta = \pi/4$ an equilibrium position?

For the next task denote the unstretched spring length simply by L_0.

(b) For which values of k is $\theta = \pi/4$ a *stable* equilibrium position?

Solution:

(a) The system is conservative, and therefore we can identify its equilibrium by requiring that $\partial V/\partial\theta = 0$. The total potential energy, comprising the spring and gravitational energies, is written as

$$V = -mgL\sin\theta + \frac{1}{2}k(L\cos\theta - L_0)^2, \tag{27}$$

so that

$$\frac{\partial V}{\partial \theta} = 0 \qquad \Leftrightarrow \qquad -mgL\cos\theta - k(L\cos\theta - L_0)L\sin\theta = 0. \tag{28}$$

By inserting the given value for the equilibrium configuration of $\theta_{\text{eq}} = \pi/4$ into the above equation and rearranging the terms to solve for L_0, we obtain

$$L_0 = \frac{mg\cos\theta_{\text{eq}}}{k\sin\theta_{\text{eq}}} + L\cos\theta_{\text{eq}} \qquad \Rightarrow \qquad L_0 = \frac{mg}{k} + \frac{\sqrt{2}}{2}L. \tag{29}$$

(b) For stability, it must hold that (for a general L_0)

$$\left.\frac{\partial^2 V}{\partial \theta^2}\right|_{\theta_{\text{eq}}} > 0 \qquad \Rightarrow \qquad mgL\sin\theta_{\text{eq}} + kL^2\sin^2\theta_{\text{eq}} - kL\cos\theta_{\text{eq}}(L\cos\theta_{\text{eq}} - L_0) > 0. \tag{30}$$

Substituting $\theta_{\text{eq}} = \pi/4$, we obtain

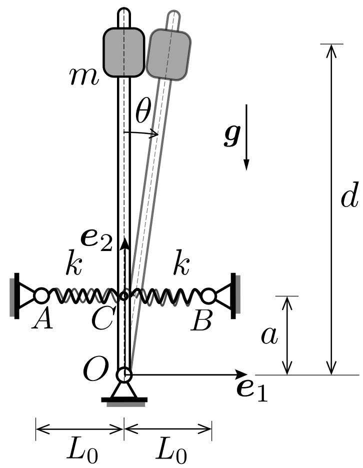
$$\frac{1}{\sqrt{2}}mgL + \frac{1}{\sqrt{2}}kLL_0 > 0 \qquad \Rightarrow \qquad mg + kL_0 > 0, \tag{31}$$

which is always satisfied for any $k > 0$.

Problem 3 (Homework): Metronome

A metronome is modelled as a massless rod, hinged at its tip O, with a particle of mass m mounted at a distance d from O. Two elastic springs of stiffness k and *zero* unstretched length support the bar at a distance a from O. When the bar is vertical, the springs are horizontal. Gravity acts downwards, as shown. Assume small oscillations.

Given: m, a, d, k, $|\theta(t)| \ll 1$, g



What is the natural frequency ω of the system?

Hint: Considering the effect of gravity can be challenging for small angles θ in this problem. To this end, you may want to expand the exact potential energy due to gravity for small angles θ about the equilibrium up to leading order in θ.

Solution: To obtain the natural frequency, we first derive the equations of motion, which can be identified by applying the Lagrange equation. The potential energy V results from gravity and the springs and can be written as

$$V = mgd\cos\theta + \frac{1}{2}k\Delta_1^2 + \frac{1}{2}k\Delta_2^2, \tag{32}$$

where θ is the angle the bar encloses with the vertical direction, while Δ_1 and Δ_2 are the magnitude of elongations of springs AC and CB, respectively. For small θ, we consider $\sin\theta \approx \theta$ and $\cos\theta \approx 1$, and thus

$$\Delta_1 = |\boldsymbol{r}_C - \boldsymbol{r}_A| = |(a\sin\theta + L_0)\boldsymbol{e}_1 + (a\cos\theta - a)\boldsymbol{e}_2| \quad \approx \quad |a\theta + L_0|, \tag{33}$$

$$\Delta_2 = |\boldsymbol{r}_C - \boldsymbol{r}_B| = |(a\sin\theta - L_0)\boldsymbol{e}_1 + (a\cos\theta - a)\boldsymbol{e}_2| \quad \approx \quad |a\theta - L_0|. \tag{34}$$

The total potential energy can hence be written as

$$V = mgd\cos\theta + \frac{1}{2}k(a\theta + L_0)^2 + \frac{1}{2}k(a\theta - L_0)^2 = mgd\left(1 - \frac{\theta^2}{2}\right) + ka^2\theta^2 + kL_0^2, \tag{35}$$

where we included the second-order terms of the Taylor series for $\cos\theta$, i.e., $\cos\theta \approx 1 - \frac{\theta^2}{2}$. Note that it is necessary to retain the $\mathcal{O}(\theta^2)$-term in this expansion to be consistent in the order of the considered approximations, i.e., to match the leading θ^2-term of the spring potential. If one instead only retains the $\mathcal{O}(1)$-term, the contribution of gravity will be lost when evaluating $\frac{\partial V}{\partial \theta}$ in the Lagrange equation.

The kinetic energy T is simply

$$T = \frac{1}{2}mv^2, \tag{36}$$

with the speed $v = d\dot{\theta}$. The kinetic energy is thus

$$T = \frac{1}{2}mv^2 = \frac{1}{2}md^2\dot{\theta}^2. \tag{37}$$

The Lagrange equation follows as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0, \quad \text{with } \mathcal{L} = T - V, \tag{38}$$

where the only non-zero terms evaluate to

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{\theta}} = md^2\ddot{\theta}, \tag{39}$$

$$\frac{\partial V}{\partial \theta} = (-mgd + 2ka^2)\theta, \tag{40}$$

so that the linearized equation of motion is obtained as

$$md^2\ddot{\theta} + (2ka^2 - mgd)\theta = 0. \tag{41}$$

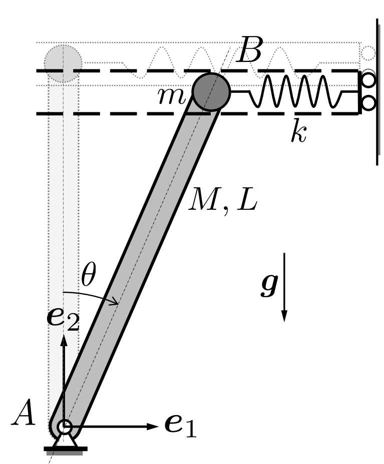
From this, the natural frequency of the system is identified as

$$\omega_0 = \sqrt{\frac{2ka^2 - mgd}{md^2}}. \tag{42}$$

Problem 4 (Homework): Bar-spring system II

A homogeneous bar of mass M and length L is hinged at point A, as shown below. A particle of mass $m = 3M$ is attached to the other end of the bar at point B, which is also connected to an elastic spring of stiffness k. The spring is unstretched when $\theta = 0$ and constrained to remain horizontal at all times. Gravity acts downwards, as shown.

Given: M, $m = 3M$, L, k, g



Under what condition for k is $\theta = 0$ guaranteed to be a stable equilibrium position?

Solution: The potential energy V, comprising both the gravitational potential of both masses and the spring energy, is given by

$$V = \frac{1}{2}kL^2\sin^2\theta + Mg\frac{L}{2}\cos\theta + mgL\cos\theta. \tag{43}$$

Taking the derivative with respect to θ, we obtain the condition for an equilibrium as

$$\frac{\partial V}{\partial \theta} = kL^2\sin\theta\cos\theta - \left(\frac{M+2m}{2}\right)gL\sin\theta = 0. \tag{44}$$

For the particular case $\theta = 0$, we have $\sin(\theta = 0) = 0$, so that all terms indeed vanish, so that $\theta = 0$ is an equilibrium solution. For stability, we must have

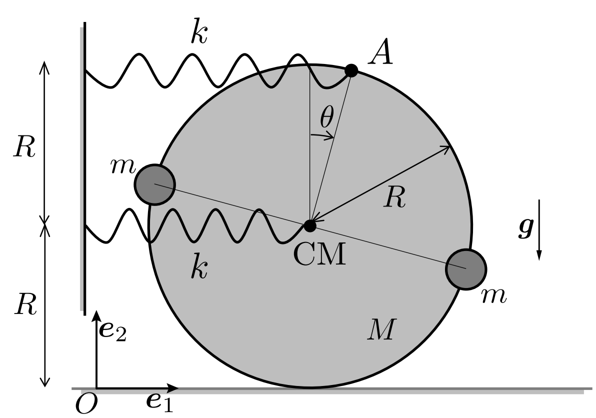
$$\left.\frac{\partial^2 V}{\partial \theta^2}\right|_{\theta=0} > 0 \qquad \Leftrightarrow \qquad k > \frac{M+2m}{2L}g = \frac{7Mg}{2L}. \tag{45}$$

This is the condition which guarantees that $\theta = 0$ is a stable equilibrium.

Problem 5 (Homework): Particle-disk system

A disk of mass $M = 2m$ and radius R rolls without slipping on a horizontal plane. Two particles of equal masses m are fixed on the disk on its perimeter opposite from each other, as shown. Furthermore, two horizontal elastic springs of stiffnesses k connect the center of mass CM of the disk and point A to the wall, as shown. Gravity acts downwards. Denote with θ the (small) rotational angle of the disk (such that both springs remain approximately horizontal). When $\theta = 0$, the two particles are at the same height R above the ground and the springs are unstretched.

Given: m, $M = 2m$, R, k, $|\theta(t)| \ll 1$, g



What is the eigenfrequency ω_0 of the system for a free vibration around $\theta = 0$?

Solution: We can find the eigenfrequency from the equation of motion of the system. The latter is conveniently obtained from the Lagrange equation, for which we need to identify the kinetic energy T

and potential energy V. (Note that one can alternatively also use linear and/or angular momentum balance, but we choose the shorter option here.) To obtain the translational component of T, we derive the velocity of CM with the velocity transfer formula (knowing the ICR on the ground) as

$$v_{\mathrm{CM}} = R\dot{\theta}, \tag{46}$$

since the disk rolls without slipping. Note that the center of mass of the system (i.e., of the disk plus the two particles) remains always at CM (the center of mass of the disk), we can evaluate the total translational component of this system.

The rotational contributions follow from the rotation of the disk and the two particles around CM, so that we can find the total kinetic energy as

$$T = \frac{1}{2}\left(\frac{1}{2}MR^2 + 2mR^2\right)\dot{\theta}^2 + \frac{1}{2}(M+2m)R^2\dot{\theta}^2 = \frac{7}{2}mR^2\dot{\theta}^2. \tag{47}$$

To obtain an expression for the potential energy, we notice that we can relate the elongation of the top and bottom springs to the rotation as $\Delta x_A = 2R\theta$ and $\Delta x_{\mathrm{CM}} = R\theta$, respectively. This gives us

$$V = \frac{1}{2}k\Delta x_{\mathrm{CM}}^2 + \frac{1}{2}k\Delta x_A^2 = \frac{1}{2}kR^2\theta^2 + \frac{1}{2}k4R^2\theta^2 = \frac{5}{2}kR^2\theta^2. \tag{48}$$

Note that the total potential due to gravity force applied to the small masses stays constant and can therefore be neglected. Correctly evaluating the Lagrange equation, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0, \quad \text{with } \mathcal{L} = T - V, \tag{49}$$

by insertion of T and V leads us to the equation of motion as

$$7mR^2\ddot{\theta} + 5kR^2\theta = 0 \tag{50}$$

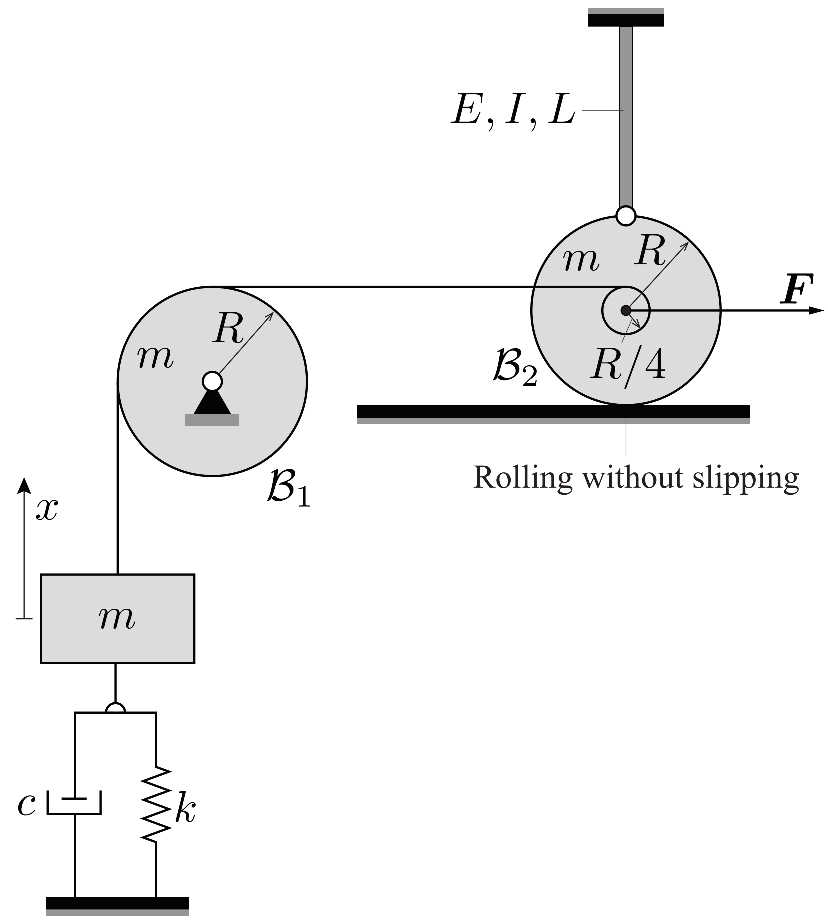
from which the eigenfrequency can directly be identified as

$$\omega_0 = \sqrt{\frac{5kR^2}{7mR^2}} = \sqrt{\frac{5k}{7m}}. \tag{51}$$

Problem 6 (Homework): Disk-block System I (2023 Final Exam)

The system below consists of a rigid block and two disks, which are connected as shown. The block of mass m, which is restricted to move vertically, is connected to an elastic spring of stiffness $k = \frac{108EI}{25L^3}$ and a velocity-proportional damper with coefficient c. The block is further connected by an inextensible string through a pulley $\mathcal{B}_1$ (on which it cannot slip and which has mass m, radius R) to a massless disk of radius $\frac{R}{4}$, which is glued onto a disk $\mathcal{B}_2$ (mass m, radius R). Disk $\mathcal{B}_2$ rolls without slipping on a horizontal plane and is further hinged at its circumference to a cantilever beam of Young's modulus E, length L, and area moment I. A horizontal force F can be applied to the center of the disk in the horizontal direction, as shown. The spring is unstretched as shown. Neglect gravity.

<u>Given:</u> m, R, c, $k = \frac{108EI}{25L^3}$, E, I, L, F



What is the equation of motion of the system in terms of the vertical position $x(t)$ of the block, which is measured from equilibrium, as shown?

Hint: the Lagrange equations can be used to solve this problem.

Solution: Denoting the clockwise rotation of the pulley $\mathcal{B}_1$ around its hinge by θ_1, and that of disk $\mathcal{B}_2$ around its point of contact with the wall by θ_2, the relations between these rotations and the DOF x are

$$\theta_1 = \frac{x}{R}, \qquad \theta_2 = \frac{4x}{5R}. \tag{52}$$

Utilizing these relations, the kinetic energy of the system is

$$T = \frac{m\dot{x}^2 + \frac{mR^2}{2}\dot{\theta}_1^2 + \frac{3mR^2}{2}\dot{\theta}_2^2}{2} = \frac{123m}{100}\dot{x}^2. \tag{53}$$

The effective stiffness of the cantilever beam is

$$k_{\text{beam}} = \frac{3EI}{L^3}, \tag{54}$$

so tgat the total potential energy of the system, considering the elasticity of the beam and the elastic spring is

$$V = \frac{kx^2 + k_{\text{beam}}(2R\theta_2)^2}{2} = \frac{6EI}{L^3}x^2. \tag{55}$$

Further, the non-conservative forces applied by the external force and the damper are jointly

$$Q^{\text{nc}} = F\frac{\mathrm{d}(R\theta_2)}{\mathrm{d}x} - c\dot{x} = F\frac{\mathrm{d}\left(\frac{4x}{5}\right)}{\mathrm{d}x} - c\dot{x} = \frac{4}{5}F - c\dot{x}. \tag{56}$$

Finally, applying Lagrange equation to the above expressions of the kinetic and potential energy as well as the non-conservative forces yields the sought equation of motion:

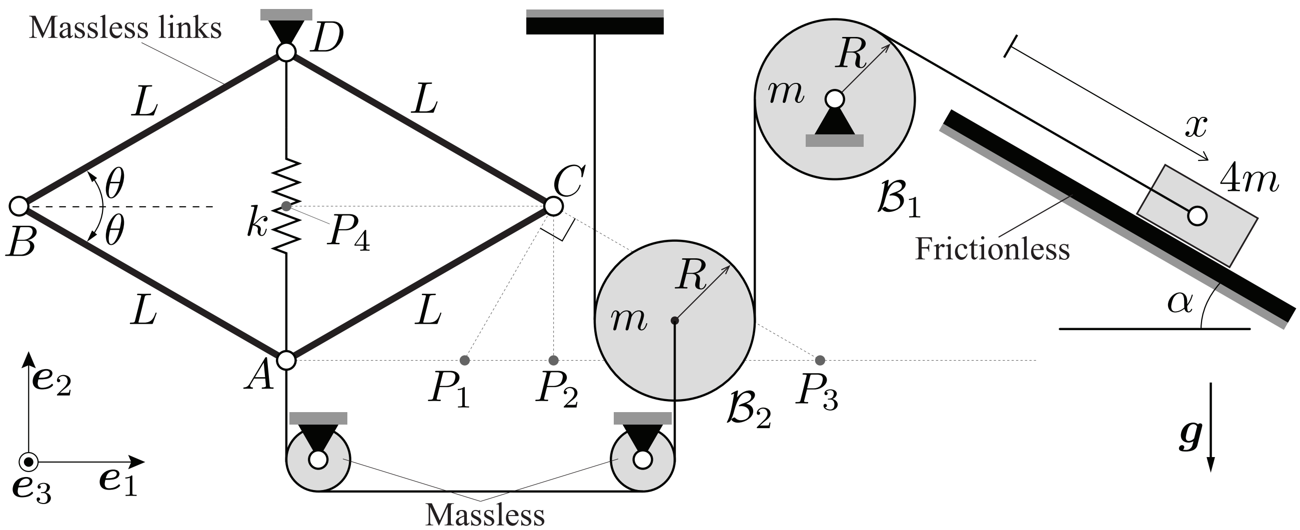
$$\frac{123}{50}m\ddot{x} + c\dot{x} + \frac{12EI}{L^3}x = \frac{4}{5}F. \tag{57}$$

Note that, if $F = \text{const.}$, then it is not a non-conservative but a conservative force. In this case, we could simply add $-FR\theta_2$ to the potential energy V instead of including F in the Q^{nc}. The final result is exactly the same.

Problem 7 (Homework): Disk-block System II (2022 Final Exam)

A block of mass $4m$ slides down an inclined frictionless slope with an inclination angle α. The block is connected to a wall by an inextensible string through a pulley $\mathcal{B}_1$ and a disk $\mathcal{B}_2$ of identical masses m and radii R. $\mathcal{B}_2$ is further connected by another inextensible string through two massless pulleys to a hinge A, which is part of the four-link mechanism $ABCD$. As shown, this mechanism consists of four identical massless rigid links of lengths L, and a spring of stiffness k, which connects hinges A and D and whose unstretched length is zero (corresponding to $\theta = 0$). Assume that the block is released from rest at $x = 0$ when $\theta = 0$. Gravity acts downwards, as shown.

Given: R, L, α, m, k, g



What is the acceleration $\ddot{x}$ of the block down the incline, when it reaches $x = \frac{mg}{k}$?

Hint: the Lagrange equations can be used to solve this problem.

Solution: Let us obtain the equation of motion from the Lagrange equation's as s shortcut (instead of formulating LMB and AMB for all components, especially since the inner forces and torques are not of interet). The kinetic energy of the system, considering the sliding motion of the block with velocity $\dot{x}$, the vertical motion of $\mathcal{B}_2$ with velocity $\dot{y}$, and the rotational motion of $\mathcal{B}_1$ (while the pulleys are massless), sums up to

$$T = \frac{4m\dot{x}^2}{2} + \frac{m\dot{y}^2}{2} + \frac{1}{2}\frac{mR^2}{2}\dot{\phi}_1^2 + \frac{1}{2}\frac{mR^2}{2}\dot{\phi}_2^2, \tag{58}$$

where ϕ_1 and ϕ_2 are the rotational angles of the pulley and the disk. Using the same kinematic variables, the potential energy of the system stemming from the gravitational forces applied to the block and the disk $\mathcal{B}_2$ together with the elastic energy of the linear spring becomes

$$V = mgy - 4mg\, x \sin\alpha + \frac{k}{2}y^2. \tag{59}$$

When the block is pulled in the x-direction, the change of length of the string connecting it to the pulley, is equal to the changes of length of both strings connected to the disk. Consequently, $y = x/2$. (This can also be seen from identifying the instantaneous center of rotation of $\mathcal{B}_2$ on the left, which shows that the center moves up by half the amount that the string on the right moves up.) Moreover, the relation between the rotation angles of both pulleys is $\phi_1 = 2\phi_2 = x/R$ for the same reason. Substituting these kinetic relations into the energy expressions yields

$$T = \frac{39m\dot{x}^2}{16}, \quad V = mg\frac{1-8\sin\alpha}{2}x + \frac{kx^2}{8}. \tag{60}$$

Applying the Lagrange equation equation to these expressions, with $\mathcal{L} = T - V$, leads to

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial V}{\partial x} = 0, \tag{61}$$

since the system is conservative. Insertion of T and V gives

$$\ddot{x} = -\frac{2kx}{39m} - 4g\frac{1-8\sin\alpha}{39}. \tag{62}$$

Finally, evaluating the above at a distance $x = mg/k$ yields

$$\ddot{x} = 2g\frac{16\sin\alpha - 3}{39}. \tag{63}$$